

Power System Analysis (NOC25-EE169)

Solutions of Assignment-2

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 12

Total mark: 12 X 1 = 12

QUESTION 1:

$$L \propto \ln \frac{d}{r'}$$

$$L_1 \propto \ln \frac{d}{r'}$$

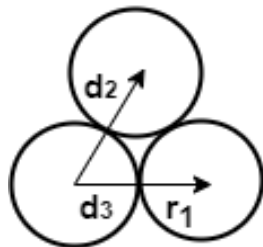
$$L_2 \propto \ln \frac{2d}{r'}$$

$$\text{As } \ln \frac{2d}{r'} > \ln \frac{d}{r'}, L_2 > L_1 \text{ but } L_2 \neq 2L_1$$

So, inductance increases but does not double.

Correct Answer: c

QUESTION 2:



In case of 3-conductor configuration,

$$d_1 = 0.7788 r_1$$

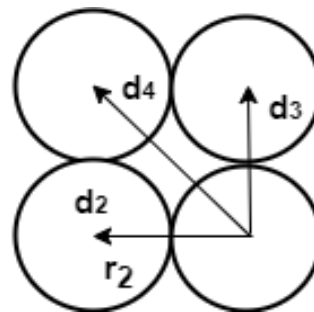
$$d_2 = d_3 = 2r_1$$

$$\begin{aligned} \text{The self GMD of the conductor} &= (d_1 * d_2 * d_3)^{1/3} \\ &= (0.7788 r_1 * 2r_1 * 2r_1)^{1/3} \\ &= 1.4605 r_1 \end{aligned}$$

In case of 4-conductor configuration,

$$d_1 = 0.7788 r_2$$

$$d_2 = d_3 = 2r_2$$



$$d_4 = ((2r_2)^2 + (2r_2)^2)^{1/2} = 2.8284 r_2$$

$$\begin{aligned} \text{The self GMD of the conductor} &= (d_1 * d_2 * d_3 * d_4)^{1/4} \\ &= (0.7788r_2 * 2r_2 * 2r_2 * 2.8284r_2)^{1/4} \\ &= 1.7229 r_2 \end{aligned}$$

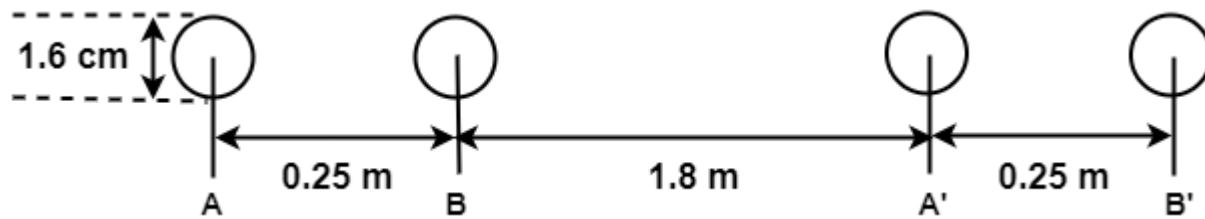
Since self GMD of the 4-conductor configuration is same as that of the 3-conductor configuration, $1.7229 r_2 = 1.4605 r_1$

$$\text{So, } r_1/r_2 = 1.7229/1.4605$$

$$= 1.1797$$

Correct Answer: b

QUESTION 3:



$$\text{Radius of each conductor} = (1.6/2) \text{ cm} = 0.8 \text{ cm}$$

$$\text{Self GMD of the line, } D_s = \sqrt{d_{AA} * d_{AB}} = \sqrt{0.7788 * 0.8 * 10^{-2} * 0.25} \text{ m} = 0.0395 \text{ m}$$

$$\text{Mutual GMD of the line, } D_m = \sqrt[4]{d_{A'B} * d_{AA'} * d_{BB'} * d_{AB'}} = \sqrt[4]{1.8 * 2.05 * 2.05 * 2.3} \text{ m} = 2.0423 \text{ m}$$

$$\text{Inductance of the split-phase, single-phase line, } L = 2 * 2 * 10^{-7} * \ln\left(\frac{D_m}{D_s}\right) \text{ H/m}$$

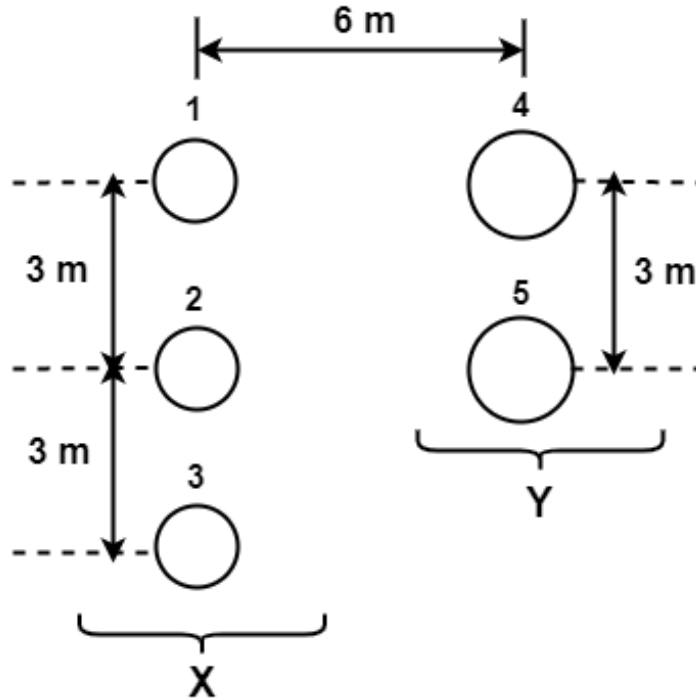
$$= 2 * 0.2 * \ln\left(\frac{D_m}{D_s}\right) \text{ mH/km}$$

$$= 2 * 0.2 * \ln\left(\frac{2.0423}{0.0395}\right) \text{ mH/km}$$

$$= 1.5782 \text{ mH/km}$$

Correct Answer: d

QUESTION 4:



The mutual GMD between set-X and set-Y conductors can be calculated as:

$$GMD_m = \sqrt[6]{D_{14} * D_{15} * D_{24} * D_{25} * D_{34} * D_{35}}$$

$$D_{14} = D_{25} = 6 \text{ m} ; D_{15} = D_{24} = D_{35} = \sqrt{3^2 + 6^2} \text{ m} = 6.7082 \text{ m} ; D_{34} = \sqrt{(3 + 3)^2 + 6^2} \text{ m} = 8.4853 \text{ m}$$

$$\text{So, } GMD_m = \sqrt[6]{6^2 * 6.7082^3 * 8.4853} \text{ m} = 6.7215 \text{ m}$$

Self GMD of set-X conductors can be calculated as:

$$GMD_s^X = \sqrt[9]{D_{11} * D_{12} * D_{13} * D_{21} * D_{22} * D_{23} * D_{31} * D_{32} * D_{33}} = \sqrt[9]{(0.7788 * 2 * 10^{-3})^3 * 3^4 * 6^2} \text{ m} = 0.2813 \text{ m}$$

$$L_X = 0.2 * \ln \left(\frac{GMD_m}{GMD_s^X} \right) \text{ mH/km} = 0.2 * \ln \left(\frac{6.7215}{0.2813} \right) \text{ mH/km} = 0.6347 \text{ mH/km}$$

Self GMD of set-Y conductors can be calculated as:

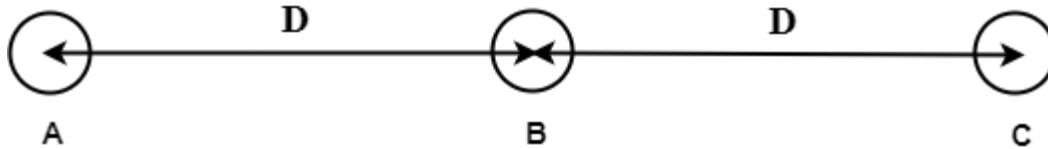
$$GMD_s^Y = \sqrt[4]{D_{44} * D_{45} * D_{54} * D_{55}} = \sqrt[4]{(0.7788 * 4 * 10^{-3})^2 * 3^2} \text{ m} = 0.0967 \text{ m}$$

$$L_Y = 0.2 * \ln \left(\frac{GMD_m}{GMD_s^Y} \right) \text{ mH/km} = 0.2 * \ln \left(\frac{6.7215}{0.0967} \right) \text{ mH/km} = 0.8483 \text{ mH/km}$$

$$\therefore L_{Total} = L_X + L_Y = (0.6347 + 0.8483) \text{ mH/km} = 1.483 \text{ mH/km}$$

Correct Answer: c

QUESTION 5:



Spacing between central conductor and outer conductor = D m

Spacing between two outer conductors = $2D$ m

Mutual GMD of the conductor, $D_m = (D \times D \times 2D)^{1/3}$
 $= 1.26D$ m

Self GMD (GMR) of the conductor, $D_s = 2 \text{ cm} = 0.02 \text{ m}$

Inductance of the line, $L = 2 \times 10^{-7} \ln(D_m/D_s) \text{ H/m}$
 $= 0.2 \ln(1.26D/0.02) \text{ mH/km}$

Inductive reactance, $X_L = 2\pi fL = 0.395 \Omega/\text{km}$

or, $2\pi \times 50 \times L = 0.395$

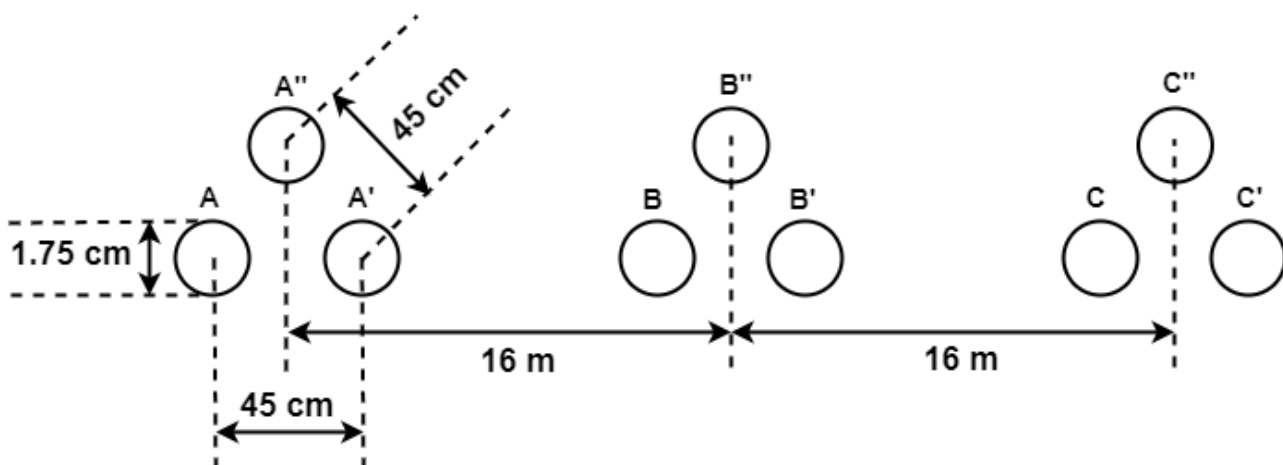
or, $L = 1.2573 \times 10^{-3} \text{ H/km} = 1.2573 \text{ mH/km}$

So, $0.2 \times \ln\left(\frac{1.26D}{0.02}\right) = 1.2573$

or, $D = 8.528 \text{ m}$

Correct Answer: a

QUESTION 6:



Radius of the conductor, $r = (1.75/2) \text{ cm} = 0.875 \text{ cm}$

Spacing between two sub-conductors of each phase, $d = 45 \text{ cm} = 0.45 \text{ m}$

Self GMD of the conductor, $D_s = (0.7788 \cdot r \cdot d^2)^{1/3}$

$$= (0.7788 \cdot 0.875 \cdot 10^{-2} \cdot 0.45^2)^{1/3} \text{ m}$$

$$= 0.1113 \text{ m}$$

Mutual GMD of the conductor, $D_m = (D_{AB} \times D_{BC} \times D_{CA})^{1/3}$

Since $0.45 \text{ m} \ll 16 \text{ m} < 32 \text{ m}$, $D_{AB} = 16 \text{ m}$, $D_{BC} = 16 \text{ m}$, $D_{CA} = (16+16) \text{ m} = 32 \text{ m}$

$$D_m = (16 \times 16 \times 32)^{1/3} \text{ m} = 20.1587 \text{ m}$$

Inductance of the line, $L = 0.2 \cdot \ln(D_m / D_s) \text{ mH/km}$

$$= 0.2 \cdot \ln(20.1587 / 0.1113) \text{ mH/km}$$

$$= 1.0398 \text{ mH/km}$$

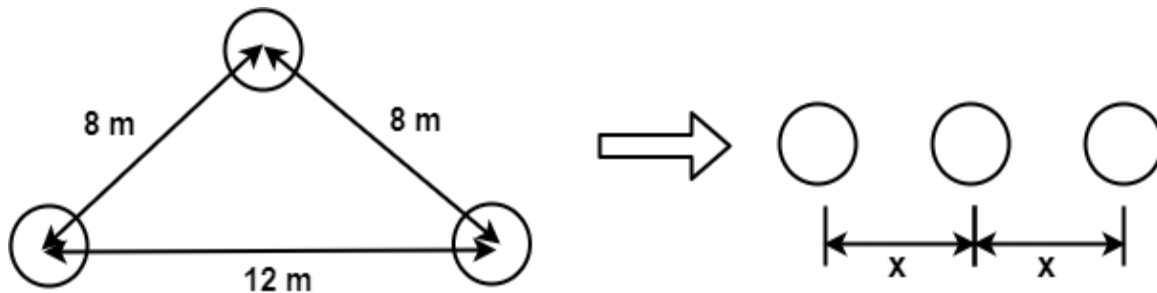
Inductive reactance, $X_L = 2\pi fL$

$$= 2\pi \cdot 50 \cdot 1.0398 \cdot 10^{-3} \Omega/\text{km}$$

$$= 0.3267 \Omega/\text{km}$$

Correct Answer: a

QUESTION 7:



In original design, mutual GMD of the conductor, $D_{m1} = (8 \cdot 8 \cdot 12)^{1/3} \text{ m} = 9.1577 \text{ m}$

In new design, mutual GMD of the conductor, $D_{m2} = (x \cdot x \cdot 2x)^{1/3} \text{ m} = x \cdot 2^{1/3} \text{ m}$

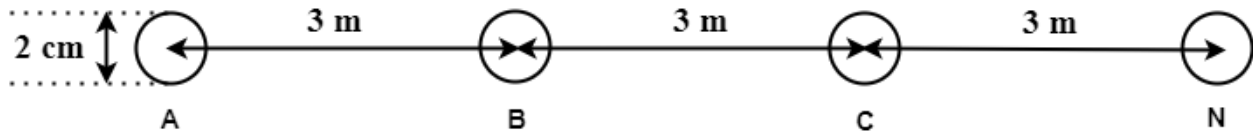
As the inductance of both the designs are same, $D_{m1} = D_{m2}$

Since $D_{m1} = D_{m2}$,

$$x \cdot 2^{1/3} = 9.1577$$

$$\text{or, } x = 7.2685 \text{ m}$$

Correct Answer: b

QUESTION 8:

Conductor radius, $r = \left(\frac{2}{2}\right) \text{ cm} = 1 \text{ cm}$

$D_{AN} = 9 \text{ m}$; $D_{BN} = 6 \text{ m}$; $D_{CN} = 3 \text{ m}$

$I_A = -20 + j26 \text{ Amp}$; $I_B = -30 + j24 \text{ Amp}$; $I_C = 50 - j50 \text{ Amp}$

Flux linkage of the neutral wire N,

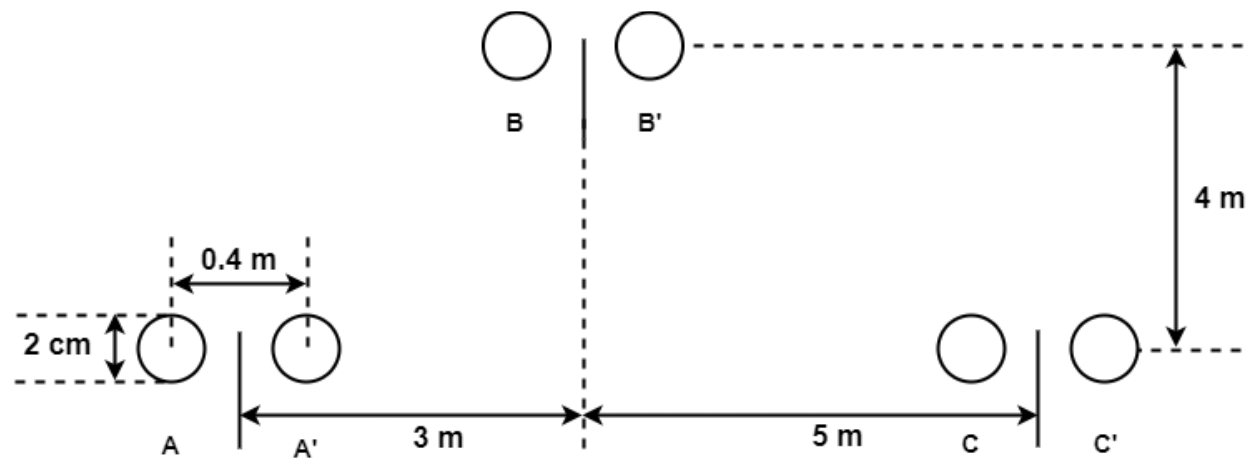
$$\lambda_N = 0.2 * \left[I_A * \ln\left(\frac{1}{D_{AN}}\right) + I_B * \ln\left(\frac{1}{D_{BN}}\right) + I_C * \ln\left(\frac{1}{D_{CN}}\right) \right] \text{ mWb-T/km}$$

$$\text{or, } \lambda_N = 0.2 * \left[(-20 + j26) * \ln\left(\frac{1}{9}\right) + (-30 + j24) * \ln\left(\frac{1}{6}\right) + (50 - j50) * \ln\left(\frac{1}{3}\right) \right] \text{ mWb-T/km}$$

$$\therefore \lambda_N = -0.2 * [2.1972 * (-20 + j26) + 1.7918 * (-30 + j24) + 1.0986 * (50 - j50)] \text{ mWb-T/km}$$
$$\text{T/km} = (8.5536 - j9.0401) \text{ mWb-T/km}$$

So, the voltage induced in the neutral wire, $V_N = j\omega\lambda_N * 35 = j2\pi * 50 * 35 * (8.5536 - j9.0401) * 10^{-3} \text{ V} = 136.844 \angle 43.416^\circ \text{ V}$

Correct Answer: c

QUESTION 9:

Radius of the conductor, $r = (2/2) \text{ cm} = 1 \text{ cm}$

Spacing between two sub-conductors of each phase, $d = 0.4 \text{ m}$

Self GMD of the conductor, $D_s = (0.7788 * r * d)^{0.5}$

$$= (0.7788 \times 1 \times 10^{-2} \times 0.4)^{0.5} \text{ m}$$

$$= 0.0558 \text{ m}$$

Mutual GMD of the conductor, $D_m = (D_{AB} \times D_{BC} \times D_{CA})^{1/3}$

Since $0.4 \text{ m} \ll 3 \text{ m} < 4 \text{ m} < 5 \text{ m}$, $D_{AB} = (3^2 + 4^2)^{1/2} \text{ m} = 5 \text{ m}$, $D_{BC} = (4^2 + 5^2)^{1/2} \text{ m} = 6.4031 \text{ m}$, $D_{CA} = 8 \text{ m}$

$$D_m = (5 \times 6.4031 \times 8)^{1/3} \text{ m} = 6.3506 \text{ m}$$

Inductance of the line, $L = 0.2 \ln (D_m / D_s) \text{ mH/km}$

$$= 0.2 \ln (6.3506 / 0.0558) \text{ mH/km}$$

$$= 0.9469 \text{ mH/km}$$

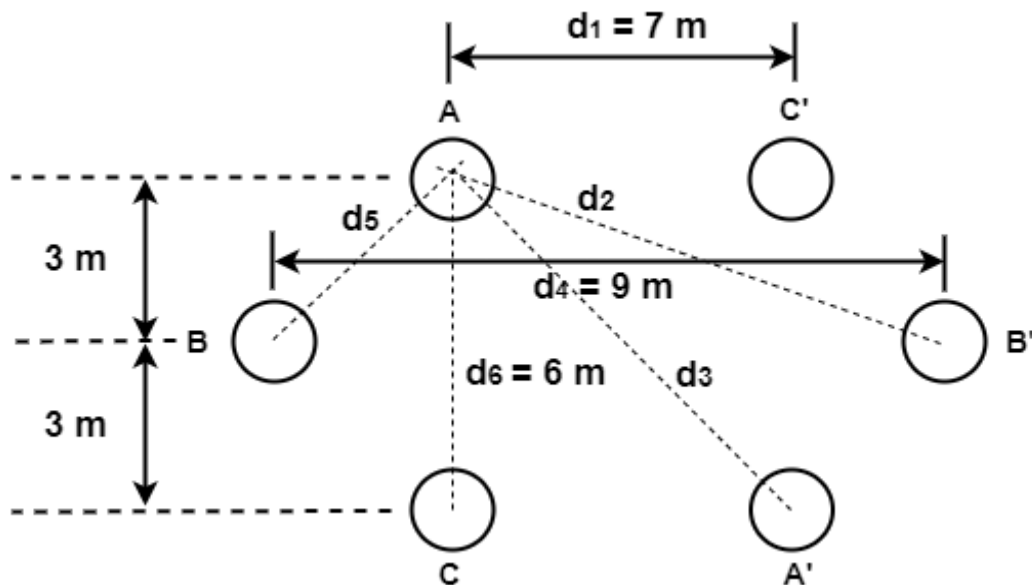
Inductive reactance, $X_L = 2\pi fL$

$$= 2\pi \times 50 \times 0.9469 \times 10^{-3} \Omega/\text{km}$$

$$= 0.2975 \Omega/\text{km}$$

Correct Answer: b

QUESTION 10:



Radius of the conductor, $r = 2.5 \text{ cm} = 2.5 \times 10^{-2} \text{ m}$

Self GMD of the conductor, $D_s = (D_{SA} \times D_{SB} \times D_{SC})^{1/3}$

$$r' = 0.7788 \times r = 0.0195 \text{ m}$$

$$d_1 = 7 \text{ m}$$

$$d_2 = (3^2 + 8^2)^{1/2} \text{ m} = 8.5440 \text{ m}$$

$$d_3 = (6^2 + 7^2)^{1/2} \text{ m} = 9.2195 \text{ m}$$

$$d_4 = 9 \text{ m}$$

$$d_5 = (1^2 + 3^2)^{1/2} \text{ m} = 3.1623 \text{ m}$$

$$d_6 = (3 + 3) \text{ m} = 6 \text{ m}$$

$$D_{SA} = (r' * d_3)^{0.5}$$

$$D_{SB} = (r' * d_4)^{0.5}$$

$$D_{SC} = (r' * d_3)^{0.5}$$

$$\begin{aligned} D_s &= (D_{SA} \times D_{SB} \times D_{SC})^{1/3} \\ &= ((r' * d_3)^{0.5} * (r' * d_4)^{0.5} * (r' * d_3)^{0.5})^{1/3} \\ &= 0.4223 \text{ m} \end{aligned}$$

$$D_{AB} = (d_2 * d_5)^{0.5}$$

$$D_{BC} = (d_2 * d_5)^{0.5}$$

$$D_{AC} = (d_1 * d_6)^{0.5}$$

$$\begin{aligned} D_m &= (D_{AB} \times D_{BC} \times D_{AC})^{1/3} \\ &= ((d_2 * d_5)^{0.5} * (d_2 * d_5)^{0.5} * (d_1 * d_6)^{0.5})^{1/3} \\ &= 5.5945 \text{ m} \end{aligned}$$

$$\begin{aligned} \text{Inductance of the line, } L &= 0.2 * \ln (D_m / D_s) \text{ mH/km} \\ &= 0.2 * \ln (5.5945 / 0.4223) \text{ mH/km} \\ &= 0.5168 \text{ mH/km} \end{aligned}$$

$$\begin{aligned} \text{Inductive reactance, } X_L &= 2\pi fL \\ &= 2\pi * 50 * 0.5168 * 10^{-3} \Omega/\text{km} \\ &= 0.1624 \Omega/\text{km} \end{aligned}$$

Correct Answer: a

QUESTION 11:

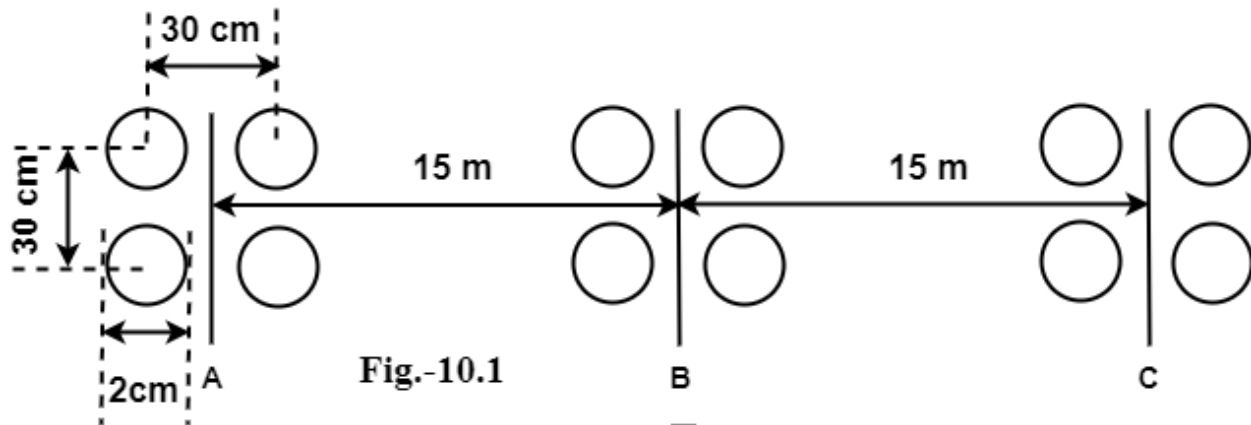


Fig.-10.1

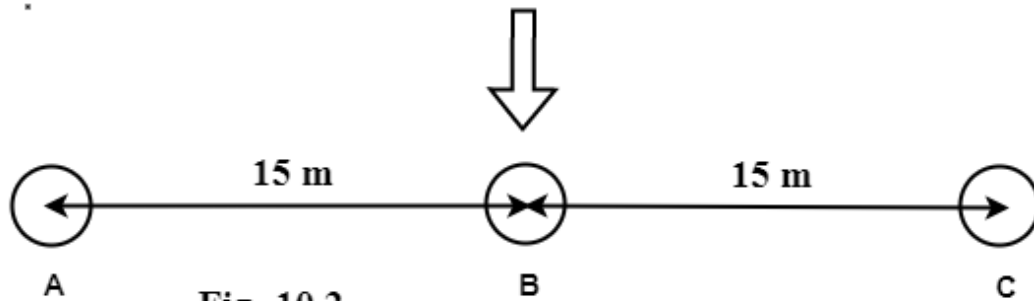


Fig.-10.2

For Fig.-10.1:

Radius of the conductor, $r = (2/2) \text{ cm} = 1 \text{ cm}$

Spacing between two sub-conductors of each phase, $d = 30 \text{ cm} = 0.30 \text{ m}$

$$\begin{aligned} \text{Self GMD of the conductor, } D_s &= (0.7788 \cdot r \cdot \sqrt{2} \cdot d^3)^{0.25} \\ &= (0.7788 \cdot 1 \cdot 10^{-2} \cdot \sqrt{2} \cdot 0.30^3)^{0.25} \text{ m} \\ &= 0.1313 \text{ m} \end{aligned}$$

Mutual GMD of the conductor, $D_m = (D_{AB} \times D_{BC} \times D_{CA})^{1/3}$

Since $0.30 \text{ m} \ll 15 \text{ m} < 30 \text{ m}$, $D_{AB} = D_{BC} = 6 \text{ m}$, $D_{CA} = 12 \text{ m}$

$$\therefore D_m = (15 \times 15 \times 30)^{1/3} \text{ m} = 18.8988 \text{ m}$$

Inductance of the line, $L_1 = 0.2 \cdot \ln(D_m / D_s) \text{ mH/km}$

$$= 0.2 \cdot \ln(18.8988 / 0.1313) \text{ mH/km}$$

For Fig.-10.2:

Let radius of the conductor is $r \text{ cm}$.

Self GMD of the conductor, $D_s = 0.7788 \cdot r \cdot 10^{-2} \text{ m}$

Spacing, $D_{AB} = D_{BC} = 15 \text{ m}$, $D_{CA} = (15 + 15) \text{ m} = 30 \text{ m}$

Mutual GMD of the conductor, $D_m = (15 \cdot 15 \cdot 30)^{1/3} \text{ m} = 18.8988 \text{ m}$

Inductance of the line, $L_2 = 0.2 \ln (D_m / D_s) \text{ mH/km}$

$$= 0.2 \ln (18.8988 / (0.7788 * r * 10^{-2})) \text{ mH/km}$$

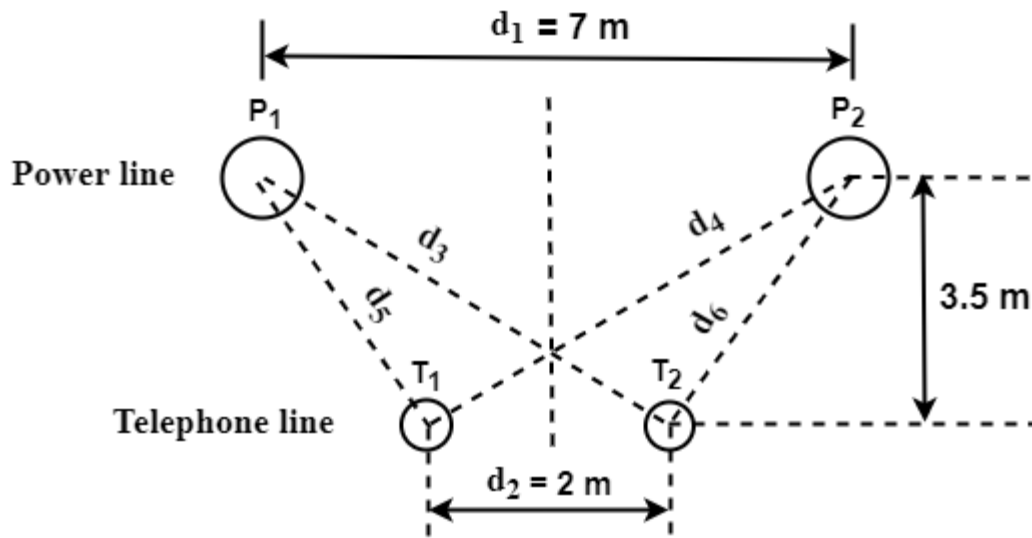
As $L_1 = L_2$,

$$0.2 \ln (18.8988 / 0.1313) = 0.2 \ln (18.8988 / (0.7788 * r * 10^{-2}))$$

or, $x = 16.86 \text{ cm}$

Correct Answer: d

QUESTION 12:



The r.m.s current in the power line, $I = 150 \text{ A}$

$$D_{P_1T_1} = d_5 = (2.5^2 + 3.5^2)^{1/2} \text{ m} = 4.3012 \text{ m}$$

$$D_{P_1T_2} = d_3 = (3.5^2 + 4.5^2)^{1/2} \text{ m} = 5.7009 \text{ m}$$

$$D_{P_2T_1} = d_4 = (3.5^2 + 4.5^2)^{1/2} \text{ m} = 5.7009 \text{ m}$$

$$D_{P_2T_2} = d_6 = (2.5^2 + 3.5^2)^{1/2} \text{ m} = 4.3012 \text{ m}$$

Flux linkage of telephone line T_1 ,

$$\lambda_{T_1} = 0.2 * \left(I * \ln \left(\frac{1}{d_5} \right) - I * \ln \left(\frac{1}{d_4} \right) \right) \text{ mWb-T/km} = 0.2 * I * \ln \left(\frac{d_4}{d_5} \right) \text{ mWb-T/km}$$

$$= 0.2 * \left(150 * \ln \left(\frac{5.7009}{4.3012} \right) \right) \text{ mWb-T/km} = 8.4519 \text{ mWb-T/km}$$

Flux linkage of telephone line T_2 ,

$$\lambda_{T2} = 0.2 * \left(I * \ln \left(\frac{1}{d_3} \right) - I * \ln \left(\frac{1}{d_6} \right) \right) \text{ mWb-T/km} = 0.2 * I * \ln \left(\frac{d_6}{d_3} \right) \text{ mWb-T/km}$$

$$= 0.2 * \left(150 * \ln \left(\frac{4.3012}{5.7009} \right) \right) \text{ mWb-T/km} = - 8.4519 \text{ mWb-T/km}$$

Total flux linkage of the telephone circuit, $\lambda_T = \lambda_{T1} - \lambda_{T2} = (8.4519 + 8.4519) \text{ mWb-T/km} = 16.9038 \text{ mWb-T/km}$

So, the magnitude of voltage induced in the telephone circuit, $V_T = \omega * \lambda_T = 2\pi * 50 * 16.9038 * 10^{-3} \text{ V/km} = 5.3105 \text{ V/km}$

Correct Answer: c

*****END*****